Computação Móvel
Mobile Computing
CM

Final Project - App

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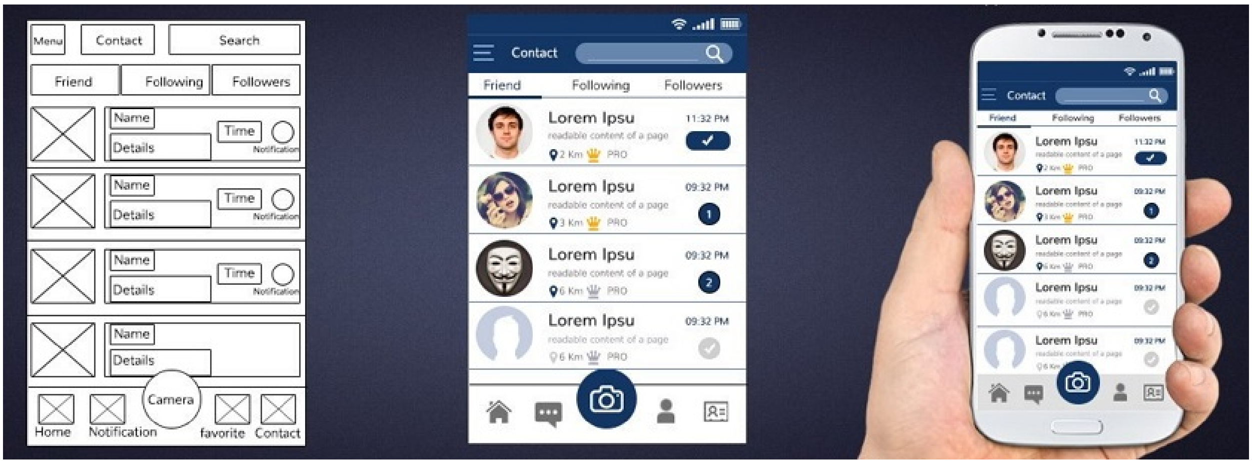
Abstract

This final project endeavors to craft an Android application grounded in a business plan, employing a development methodology centered around enhancing user experience.

Deadline:

Proposal: Mar 27^{th}, 2026
Concept: May 10^{th}, 2026
Pre-production: May 24^{th}, 2026
(Final) Market version: May 31^{st}, 2026



Android Application

March 27^{th}, 2026

Contents

1 Introduction and delivery dates

This kind of project aims to design, implement and evaluate an application for mobile devices with the Android operating system. The development process should follow the methodology centered on the user experience, which has the phases of:

1. Proposal (preliminary approval phase);
2. Concept;
3. Pre-Production;
4. Production; and
5. Post-Production.

The application should have the following:

1. at least 4 main screens;
2. use multimedia elements (images, videos, audios, etc.);
3. state, which should be saved locally and remotely (Firebase), and private and shared state, the last one shared between other all users or just with some group(s) of users;
4. a business plan with one key feature to engage users to pay a certain amount (assume 1€) per month to have access to that feature, and the perspective of having 10 million users.

a business plan from the beginning. The app concept should be conceived to have at least one key feature to engage users to pay a certain amount (assume 1€) per month to have access to that feature. An example is an App Beach, used to register beach access, where payable users can have access to history data. All users can reserve places on beaches, but only paid users can see the state of the beaches in the last days (1 week).

The application must use the Firebase platform to **save the state of the application** and share it between users. Students must be proactive and self-sufficient to gather all the necessary information to use this platform through on-line information (documentation, examples, and forums).

Students must deliver a fully functional marked-ready version of the application and evaluate it in terms of usability, functionality, and business model.

Summary of deliveries and final assessment, and their deadlines:

1. Concept phase documents and artifacts: **May 10th, 2026**;
2. Pre-production phase documents and artifacts: **May 24th, 2026**;
3. Final report and the market version of the app: **May 31th, 2026**; and

4. Final assessment (oral discussion): **until July 14th, 2026**.

2 Application Grading and Features

This chapter begins by defining the application development guidelines, then outlines the minimum requirements, and concludes with a description of the expected additional features.

2.1 Application development guidelines

The developed application will be **classified** according to the following **guidelines**:

- Application **based on an existing application**
  - Minimum features graded to **13**/20
- Application **based on an existing application**, but with **significant variations**
  - Minimum features classified to **16**/20
- **Original application**
  - Minimum features graded to **18**/20
- **Extra features** add up to 1/20 for each feature, **over minimum requirements grade**
- Features are evaluated in terms of their originality, complexity, and implementation quality

2.2 Application minimum requirements

The application **minimum requirements** are as follows:

- Code from scratch - you should build all the code;
- Support users **login**;
- Save application state using **Firebase**;
- Have, at least, **3 significant activities**, with the use of passing and returning values;
- Have screens/pages with small sized images, large sized images and at least **one animation**; and

- Have **start menu**, **help**, **about** (with author identification and photo), and **settings** (account information) and **multilingual** support for Portuguese, English, and one more language (use a translator).

2.3 Application extra features

The application **extra features** may include:

- **AI (LLM, VLM, MMLM, ...) related** functionalities;
- Multimedia content: video, sound, more animations (of different types); or
- Google cloud functions or other relevant technology.

An extra feature may be a group of requisites, not just a single item. Any additional feature must be approved previously by the teacher.

3 Development Methodology and Scheduling

The application should be designed and implemented with a focus on user experience and a business model. Students should validate the application and user interface in class with the teacher (mandatory!).

The application development should comply with the Development Methodology.

3.1 Concept phase - application concept

This phase aims to define the concept of **application** and its main artifacts and to study and test the new technologies that will be used. For this phase, it is required:

- Define the **name** and **general concept** of the application;
- Define *app* main characteristics (entities) and functionality;
- Define *app* main audience/users;
- Identify at least one **key feature** that will engage users to pay a certain amount (1€) per month to have access to that feature;
- Make *app* **wire-frame**;
- Build *app* navigation map (wire-frames and navigation);
- Build the **Application Concept Document**(**ACD**) with 2 to 4 pages;
- Get **teacher approval** for *app* concept, business model, and extra features - **this is mandatory** (without teacher approval students cannot proceed);

- Define the Entity-Association diagram (without attributes) for the *app* database to be used;
- Create a **Firebase** project (https://firebase.google.com) and implement a Firebase Android client test application with user login using Firebase User Authentication with Email/Password and, optionally, as an extra, with another third-party authentication provider such as Google, Facebook or Twitter (worth 1/20 points);
- Make the changes to enable the **Firebase Android client** test application to save and get text and images;
- If the *app* will use other technologies not tested in the classes, the student should present a minimal report with an initial test (like a "Hello World" with caffeine) of each of them; and
- Build a **report** with all these points.

The Concept Phase documents and artifacts should be delivered until **May 10th, 2026**.

3.2 Pre-production phase - application details and initial prototype

This phase aims to define all the details of **applications** and produce an **initial functional prototype**. For this phase, it is required:

- **Make user evaluation**, asking 4 users (minimum). After a short introductory explanation, ask them to see the wire-frame layouts and read the concept document and then to share their feedback, answering these questions:
  - What is the purpose of the application?
  - Was it easy to get the application concept?
  - Do you think it is appropriate for the target audience?
  - What do you think of the application in general?
  - Do you like it?
  - Would you use it?
  - Do you think there is an unsuitable element?
  - What would you like to see changed is the application?
  - How much are you willing to pay (monthly) to have access to payable features?
  - What are the payable features you would like to be added and you would pay with a smile?
- Improve the application concept and details, based on users feedback
- Design *app* **mock-ups**, using a prototyping tool of your choice
- Define the *app* user profiles

- Define the **full entity-association diagram** for the Firebase database
- Build the **Application Design Document** (ADD)
- Build a **minimal *app* version** (prototype) with all existing activities, with minimum and rough elements, only to allow navigation between activities. Use hard-coded (simulated) data if needed.

The Pre-production phase documents and artifacts should be delivered until **May 24^{th}, 2026**.

Students should finish the first two phases as soon as possible, in order to have more time for the final phases.

3.3 Production phase - pre-market version

This phase aims to focus on the **development of the *app*** and should end with a **pre-market *app* version**. For this phase, it is required:

- Complete the layouts
- Build *app* auxiliary classes to handle existing data
- **Build the *app* pre-market version**, which should support all *app* main aspects
- Make **production usability evaluation** with at least 4 users (not from the concept ones), with these questions:
  - What are the actions you can do in the app?
  - Enumerate the app screens?
  - On which screens did you spend the most time?
  - What screens did you skip?
  - Did the navigation flow as expected?
  - Would you like to change the navigation flow?
  - Do you think the app is more difficult to use?
  - Did you find a situation that frustrated you?
  - Which parts of the app would you like to see improved? How?
  - How much are you willing to pay (monthly) to have access to payable features?
  - What are the payable features you would like to be added and you would pay with a smile?
- **Refine** *app* if necessary.

3.4 Post-production - market version and final report

This phase aims to **finish *app* development** and end with a **ready-to-market *application* version**. For this phase, it is required:

- Add *app* **final details** (multi-language, help, about, settings, ...)
- Make **post-production usability evaluation**, repeating the production questionnaire with at least 6 users (not the same that already answered)
- Create the **final report** with all the documents

4 Final Report

The final report is mandatory and should include the following:

- introduction to the work
- the entire application development process
- initial wire-frame diagrams
- initial mock-ups
- final screenshots
- final full entity-association diagram of database
- final full data-base schema (no data)
- final full UML class diagram
- results obtained
- discussion of the most important issues
- conclusions

It should also include, in the appendices, all the documents produced throughout the application development (e.g., the concept document), which should be referenced in the report chapters as outcomes of the work, at their time.

5 Submission and Evaluation

The submission of the work includes the report in pdf and the source code of the application and should be done: through the Moodle platform if possible; if not (due to size limitations), through the OneDrive. If OneDrive is used, students must also send a zip, in Moodle, containing the link to the OneDrive zip file and the report (PDF), if possible.

The final report and the version of the app market must be delivered by **May 31th, 2026**.

The final evaluation of the work will be done after this delivery.

The final grade of the curricular unit includes all the grades of the tutorials performed during the semester and the final project (final product and report) and is obtained in an oral discussion.

Appendix: Wire-frames, mock-ups and prototypes

A **wire-frame** (or wireframe) is a rough **sketch** about how a website/app/game will look like. Usually is conceived in black and white and is focused on contents, not on visualization details. It should describe the existing screens, their contents, and navigation details. It should be rough, to avoid waste time in details that could reveal to be not suitable.

Figure 1: Wire-frame.

A **mock-up** (or mockup) is a high fidelity wire-frame. It focuses on visual aspects, such as colors, shapes, effects, spacing, text and form style, and navigation. It should offer a complete visual guide for the prototype phase.

A **prototype** is a real implementation of the product, but in an early stage. It should contain only the main use cases. Then, it should evolve to contain all of them.

Note: information and images from: www.mockplus.com and /www.fiverr.com.

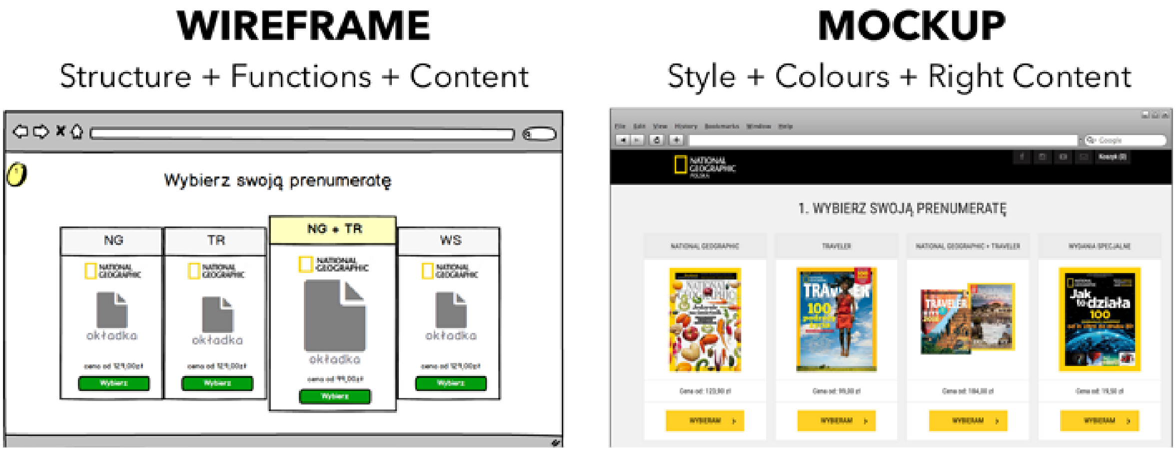


Figure 2: Wire-frame and Mock-up.

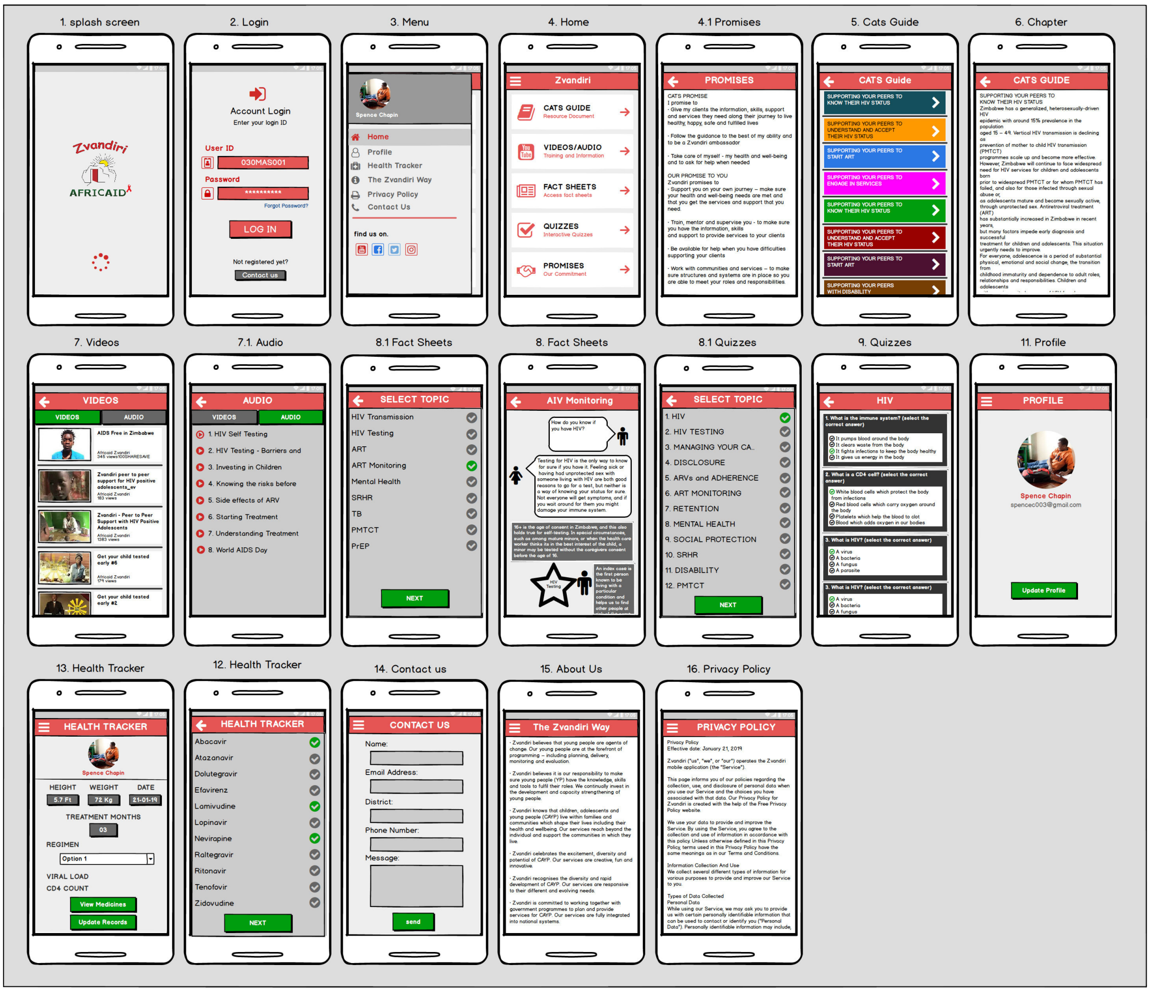


Figure 3: App full Mock-up.

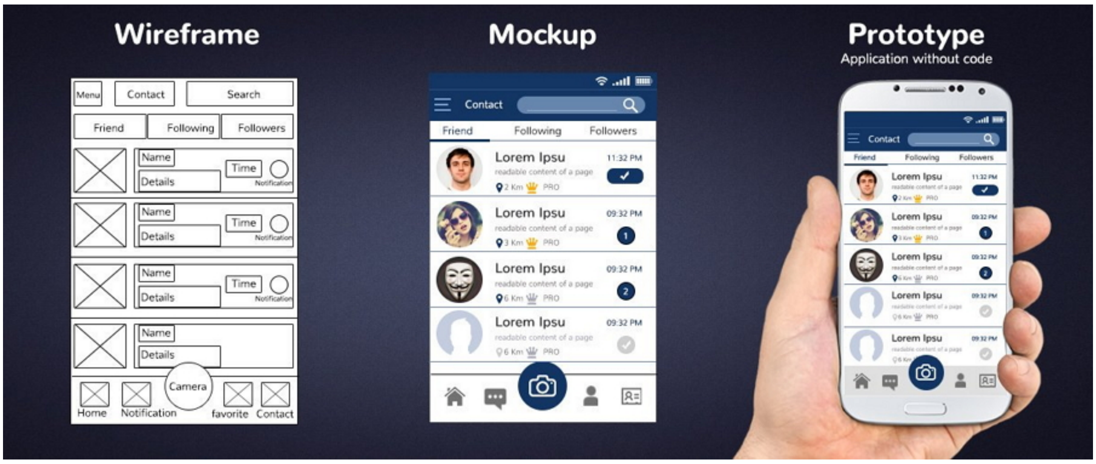


Figure 4: Wire-frame, Mock-up and Prototype.